THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Yes	Digit sum 18 is divisible by 9
2	**(d)** No	Digit sum 13 is not divisible by 3
3	**(a)** Yes	Difference of sums of alternate digits
4	**(c)** 7	Digit sum must be a multiple of 9
5	**(a)** 36	Smallest common multiple
6	**(c)** 48	Smallest common multiple
7	**(c)** 30	LCM
8	**(d)** even	Property
9	**(a)** neither prime nor composite	By convention
10	**(a)** 100	Ends in two zeros

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A number is divisible by 3 if the sum of its digits is divisible by 3 (2).	2
12	The difference between the sums of digits at odd and even places is 0 or a multiple of 11 (2).	2
13	Last two digits 48 are divisible by 4, so yes (2).	2
14	Last three digits $392 = 8 \times 49$, so yes (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Digit sum condition (2); digits: 1, 4, 7 (1).	2+1
16	Digit sum condition (2); digits: 4 (1).	2+1
17	Digit sum condition (2); digits: 3 (1).	2+1
	OR: Digit sum condition (2); digits: 0, 3, 6, 9 (1).	2+1
18	Digit sum condition (2); digits: 8 (1).	2+1
	OR: Digit sum condition (2); digits: 0, 3, 6, 9 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Last digit; last two digits; last digit 0/5; last three digits (3). (ii) Appropriate checks (2).	3+2
	OR: (i) $84 = 2^2 \times 3 \times 7$, $126 = 2 \times 3^2 \times 7$; HCF 42, LCM 252 (3). (ii) $42 \times 252 = 10584 = 84 \times 126$ (2).	3+2
20	(i) LCM $= 360$ s (2). (ii) 9:06 a.m. (1). (iii) 11 times (2).	2+1+2
	OR: (i) $11(A + B) = 121$, $A + B = 11$, e.g. 29 and 92 (2). (ii) $11(a + b)$ (3).	2+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 5: **Number Play** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Yes	Digit sum 15 is divisible by 3
2	**(a)** Yes	Digit sum 18 is divisible by 9
3	**(d)** Yes	Difference of sums of alternate digits
4	**(b)** 2	Digit sum must be a multiple of 3
5	**(d)** 40	Smallest common multiple
6	**(d)** 36	Smallest common multiple
7	**(a)** both 2 and 3	6 = 2 × 3 (coprime)
8	**(c)** 6174	Repeated rearranging and subtracting
9	**(a)** 2	Only even prime
10	**(d)** 36	9+9+9+9

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Numbers whose HCF is 1 (1); 8 and 15 (1).	1+1
12	Factors: $18 = 2 \times 3^2$, $48 = 2^4 \times 3$ (1); HCF $= 6$ (1).	1+1
13	$\text{LCM} = 24$ (2).	2
14	Reads the same both ways (1); 121 and 4554 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Prime factorise each (1); take the highest powers (1); LCM $= 180$ (1).	1+1+1
16	Prime factorise each (1); take the highest powers (1); LCM $= 120$ (1).	1+1+1
17	Prime factorise each (1); take the highest powers (1); LCM $= 300$ (1).	1+1+1
	OR: Prime factorise each (1); take the highest powers (1); LCM $= 90$ (1).	1+1+1
18	Prime factorise each (1); take the highest powers (1); LCM $= 210$ (1).	1+1+1
	OR: Prime factorise each (1); take the highest powers (1); LCM $= 240$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) 954−459=495 (2). (ii) Reaches the constant 495 (1). (iii) 100−001=099; 990−099=891; 981−189=792; 972−279=693; 963−369=594; 954−459=495 (2).	2+1+2
	OR: (i) LCM 99; $1000 \div 99$ gives 1089 (2.5). (ii) $9999 = 99 \times 101$ (2.5).	2½+2½
20	(i) HCF $= 36$ (2). (ii) 4 red and 5 blue rows (2). (iii) Largest equal group (1).	2+2+1
	OR: (i) $10^k \equiv 1 \pmod 9$ so number ≡ digit sum (3). (ii) Digit sum 18, divisible by 9 (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.